THE MALAWI NATIONAL EXAMINATIONS BOARD

2000 MALAWI SCHOOL CERTIFICATE OF EDUCATION EXAMINATION

MATHEMATICS

Subject Number: M131/II

Time Allowed : 2h 30 mins
1:30 - 4:00 p.m.

Tuesday, 19 December

PAPER II
(100 marks)

Instructions

1. **This paper contains 6 pages. Please check.**
2. Answer **all** the **six** questions in Section A and any **three** questions from Section B.
3. The maximum number of marks for each answer is indicated against each question.
4. Mathematical tables, graph paper and answer books are provided.
5. Used graph paper and/or supplementary sheets **must** be tied together inside the answer book with a string.
6. All working must be clearly shown: it should be done on the same sheet as the rest of the answers.
7. Use of electronic calculators is **not** allowed in this paper.

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Turn over

Section A (55 marks)

Answer **all** the **six** questions from this section.

1. (a) Subtract 2a - 7b from the sum of 4a + 2b and 4a - 11b. (3 marks)

   (b) A pail is $\frac{3}{5}$ full of beans. When 8 full tins of beans are added, it is $\frac{5}{7}$ full. How many more tins must be added to fill the pail completely? (5 marks)

2. (a) Simplify $\frac{2}{\sqrt{2}} - \frac{1}{5\sqrt{2}}$, leaving your answer with rational denominator. (3 marks)

   (b) In figure 1, PQRS is a circle. QPT and RST are straight lines.

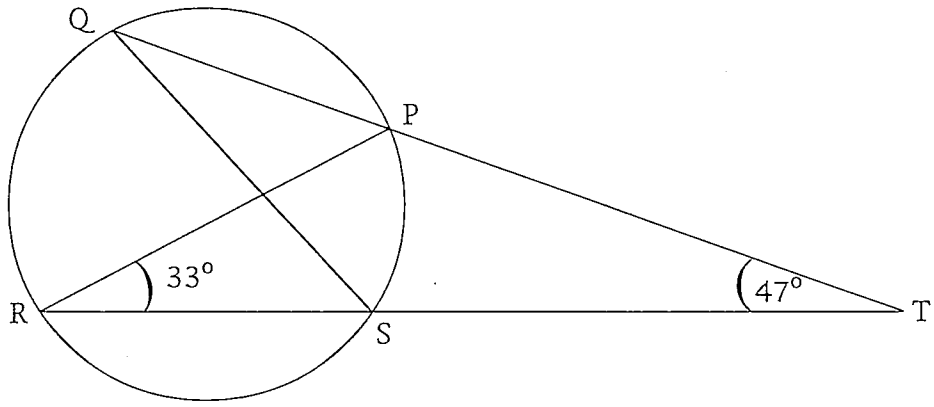

Figure 1

If angle PRS = 33° and angle STP = 47°, calculate angle QSR. (3 marks)

3. (a) The table below shows scores obtained by a student in an end of term examination.

SUBJECT	STUDENT'S SCORE	TOTAL MARKS
Mathematics	16	20
Science	42	50
English	28	40
Bible Knowledge	60	80

   (i) Calculate the mean percentage mark.
   (ii) When home economics mark is included, the mean percentage is 77. What was the student's score in home economics if the test was marked out of 25? (8 marks)

   (b) Make t the subject of the formula

$$y = \sqrt{\frac{a - t}{t + b}}$$

(6 marks)

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4. (a) In figure 2, PAT and CT are tangents to the circle ACB.

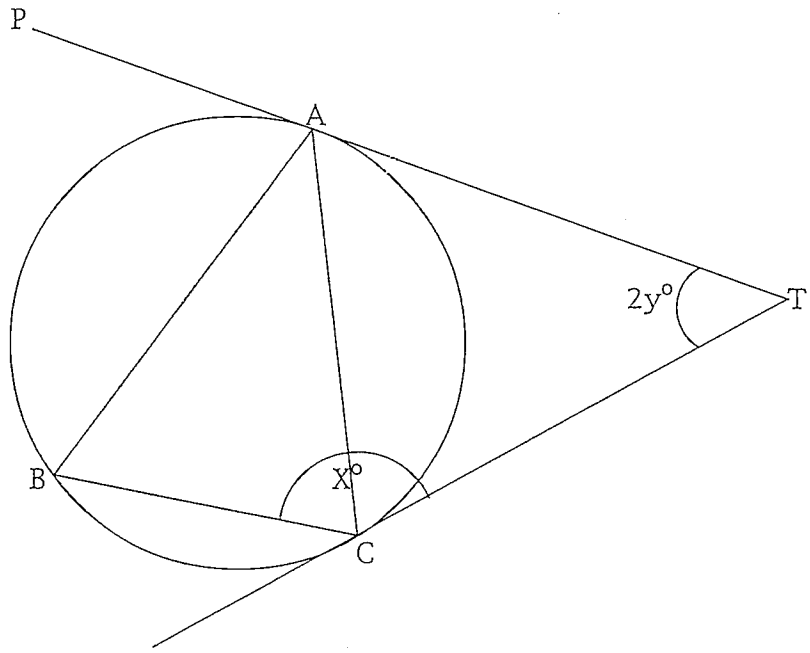

Figure 2

Given that angle BCT = x° and angle CTA = 2y° , express angle PAB in terms of x° and y°. (6 marks)

(b) The ages of 10 students at a certain school are:

15, 12, 20, 18, 23, 16, 20, 24, 20, 15.

Find (i) the modal age

(ii) the median age. (3 marks)

5. Prove that chords of a circle which are equidistant from the centre are equal. (7 marks)

6. (a) Solve the equation $2a^2 + 4a = 3$, giving your answer correct to 3 significant figures. (7 marks)

(b) Calculate the total surface area of a solid circular cylinder whose diameter and height are 2.58 cm and 4.27 cm respectively. (Take $\pi = 3.14$) (4 marks)

Section B (45 marks)

Answer any **three** questions from this section

7. (a) A man at the top of a building observes a car moving on a straight road towards the building. The initial position of the car was P and after 10 seconds, the car was at Q. The angles of depression of P and Q were 40° and 70° respectively. If the height of the building is 50 metres, calculate the distance between P and Q. (5 marks)

(b) In figure 3, ABCD is a parallelogram. AB is produced to E such that AB = BE. AHC, BGC and DHGE are straight lines.

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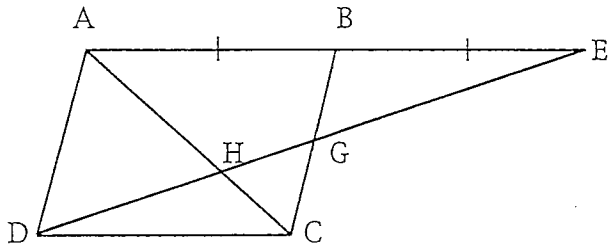

Figure 3

Prove that:

(i) G is the mid point of BC.
(ii) triangle HCD is similar to triangle HAE.
(iii) CH = ½ HA (10 marks)

8. (a) The total bus fare for x passengers is K288.00. If one of them had lost his money, the others would have contributed K4.00 each towards his fare. How many passengers were there? (7 marks

(b) Villages A and B are 120 km apart. A cyclist travelling at 20 km/h leaves village A at 6.00 a.m. for Nakala Market 20 km away towards Village B. After resting for 30 minutes at the market, he returns at the same speed and reaches Village A at 9.30 a.m.

A motorist leaves Village B for Village A at 7.00 a.m. and travels at an average speed of 80 km/h. She stops for 15 minutes at the market and proceeds to Village A where she arrives at 9.00 a.m.

Taking a scale of 2 cm to represent 10 km on the vertical axis and 2 cm to represent 30 minutes on the horizontal axis, draw the graphs on the same axes representing the cyclist's and motorist's journeys.

Use your graphs to answer the following questions:

(i) At what time was the motorist 80 km away from Village A?
(ii) How far from the market was the motorist when the cyclist was leaving Nakala? (8 marks)

9. (a) The table below shows values of x and y of the function

$y = 2x^2 - 3x - 7.$

x	-2	-1	0	1	2	3	4
y	7	-2	-7	-8	-5	2	13

Continued →

Using a scale of 2 cm to represent 1 unit on the x - axis and 2 cm to represent 5 units on the y axis,
draw the graph of $y = 2x^2 - 3x - 7$.

Use your graph to solve the following equations:

(i) $2x^2 - 3x - 7 = 0$
(ii) $2x^2 - 5x - 6 = 0$ (10 marks)

(b) In a regular polygon ABCD...., three consecutive sides are AB, BC and CD. The angle ABD is 9 times angle CBD.
Calculate the size of each interior angle. (5 marks)

10. (a) The total cost of building a house in 1990 was K90 000.00 which was made up of materials, wages and administrative expenses, in the ratio 4:5:1 respectively. By 1993 the cost of materials had risen by 25%, wages by 20% and administrative expenses by 10%.

Calculate:

(i) the new cost of building a similar house;
(ii) the overall percentage increase for the total cost of the house.
(8 marks)

(b) In figure 4, AB is a diameter of a circle and P is a point on the circle such that PA is less than PB.

PM is perpendicular to AB and PN bisects angle MPB.

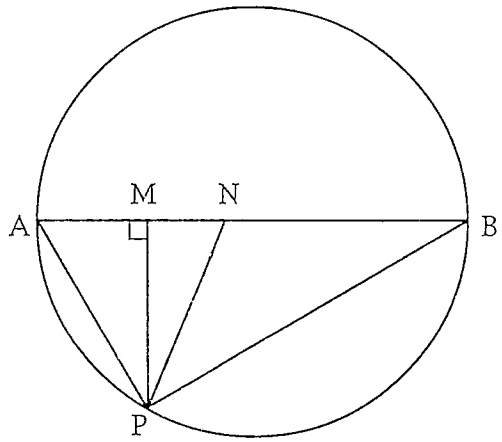

Figure 4

Prove that:

(i) angle APM = angle ABP

(ii) AP = AN. (7 marks)

11. (a) The effort required to raise a load is partly constant and partly proportional to the load. The effort necessary for a load of 8 newtons is 6 newtons and for a load of 12 newtons is 8 newtons. Find the effort required to raise a load of 16 newtons. (7 marks)

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(b) FN is an altitude of triangle FBC. Prove that

$FB^2 - FC^2 = BN^2 - CN^2$ (3 marks)

(c) PQRS is a parallelogram. F is any point on PQ such that FR = QR.

Prove that P, F, R, S are concyclic points. (5 marks)

12. (a) In a geometrical progression, the sum of the second and third terms is 9. If the seventh term is eight times the fourth, find:

(i) the first term;

(ii) the common ratio. (6 marks)

(b) Figure 5 is a triangular prism with DE = 3 cm, EF = 2 cm, DF = 4 cm and BE = 12 cm.

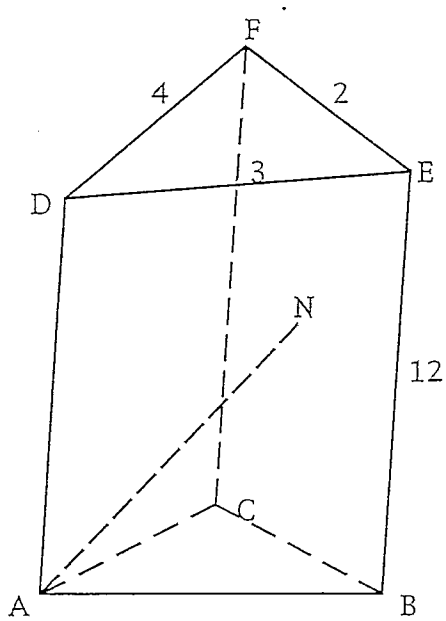

Figure 5

If N is the point where diagonals of the face FEBC meet, find the angle which NA makes with the plane ABC. (9 marks)

END OF QUESTION PAPER

NB: This paper contains 6 pages